

A Noninvasive Machine Learning Model Using a Complete Blood Count for Screening of Primary Vitreoretinal Lymphoma

Corresponding Author: Dr Shengjie Li

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This manuscript describes the development and validation of a random forest (RF) machine learning-based screening model, for the prediction of primary vitreoretinal lymphoma (PVRL) from six input features derived from complete blood count (CBC) data. The RF model achieved AUC=0.85 in the discovery/test set, and was further evaluated on a large prospective cohort of some 100,000 subjects, and a community cohort of over 500,000 subjects.

The potential of an accurate noninvasive screening tool for PVRL from commonly-encountered CBC data is of interest, especially given the lack of a standardized diagnostic protocol for PVRL. Some issues might be considered:

1. In the Introduction section, it is claimed that there is no standardized diagnostic protocol for PVRL. The choice of the 2016 WHO classification (as described in the Diagnostic Criteria for PVRL subsection in Line 147) might thus be clarified further. Is the 2016 WHO classification biopsy protocol considered the gold standard for PVRL diagnosis then?
2. In the Study design and participants subsection, it is mentioned that healthy controls were recruited from the health examination centres of various hospitals. It might first be clarified as to whether these were the same hospitals that the PVRL patients were recruited from. Then, since the study is described as case-control, it might be clarified as to how controls were matched to cases (e.g. by age, gender, etc.), and whether such matching was one-to-one or one-to-many.
3. Related to the above, a flowchart detailing the recruitment and exclusion for both cases and controls for the case-control multicentre study should be included if possible (as was done for the external cohorts in Figure 5).
4. In the Study design and participants subsection, it is then stated that PVRL screening was restricted to individuals over 40 years of age, to enrich prevalence within the study cohort. However, given that PVRL has extremely low prevalence to begin with, would this not unnecessarily limit the applicability of the developed model for the younger (<40 year-old) population? This exclusion criteria might thus be further justified.
5. In the Diagnostic Criteria for PVRL subsection, it is stated that "PVRL was diagnosed based on a positive pathological finding from vitreous biopsy". Any detailed methodology and quantitative criteria leading to said positive pathological finding, might be described.
6. In the Model development subsection, it is stated that 29 input features were available to develop the diagnostic models. These 29 features might be provided, possibly in the supplementary material (are they in Table S2)?
7. In the Model development subsection, it is stated that grid search and manual finetuning were used to determine the hyperparameters for each model. From the supplementary material, 5-fold cross-validation was performed (Figure S1), which implies possibly 5 different sets of optimal hyperparameters (one for each fold). It might thus be further clarified as to how the hyperparameters were derived (e.g. are the 5-fold results shown, from the set of hyperparameters with the optimal mean performance over all folds?)

8. In the Feature selection and model explanation subsection, it is stated that sample features were selected in the discovery cohort with SHAP, and this narrowed the model from 29 features to 1 feature. However, the abstract claims that the RF model includes 6 features (also in Figure 2). The claim of 1 feature might thus be clarified.

9. The final choice of 6 features (and not for example 5 or 7 features) might be justified. Was some feature selection methodology applied (e.g. progressively adding features to the models in order of SHAP value, then evaluating the performance on the discovery set)? This might be described in detail.

10. For the hospital-based and community-based validation, it is assumed that the gold standard biopsy-based ground truth for PVRL was not available for all patients. The analysis of false negatives (i.e. PVRL patients judged to be low-risk and thus not examined in detail, especially for the community-based group) might thus be justified further.

(Remarks on code availability)

The GitHub code appears to be standard implementations of various common machine learning models in Python/SkLearn.

Reviewer #2

(Remarks to the Author)

This interesting study aims to develop a machine learning-based screening model for detection of PVRL using particular selected complete blood count parameters, i.e., monocyte percentage, Platelet Large Cell Ratio (PLCR), monocyte count, HG and basophil count. I congratulate the authors for their very thorough study, demonstrating multiple validation levels across both inpatient and outpatient settings. The results show that a six-feature-based random forest machine learning model is capable of predicting patients with vitreoretinal lymphoma. Furthermore, the study also mentions a comparison of patients with VRL with those with uveitis and vitreous opacities. It would be very interesting to know the specific diagnostic entities for the latter, because it's one thing to consider easily differentiated clinical uveitis (herpetic uveitis), but it's quite another to consider uveitis from sarcoidosis or endogenous endophthalmitis as a differential diagnosis. For these patients, it has been demonstrated that radiomic analysis of anterior segment OCT images is a non-invasive diagnostic aid (Gozzi et al. Diagnostics 2023).

One aspect that needs to be emphasized, however, is that this machine learning method based on these CBC parameters must ALWAYS be confirmed by a biopsy, i.e., vitrectomy.

(Remarks on code availability)

I was not able to install and run the code

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

We thank the authors for addressing our previous comments. Some minor issues might be considered:

1. In the response to Point 2, it is confirmed that a one-to-one matching strategy was applied. In this case, the range for age being matched on might be stated - did the matching have to be exact, or within e.g. 2 years?

2. In the response to Point 5, it might be briefly clarified as to whether the diagnosis of PVRL from vitreous biopsy followed any established standards, with appropriate references if available.

3. In the response to Point 7, it is stated that nested 5-fold cross-validation was conducted. It might be clarified whether this nested cross-validation was performed only on the discovery cohort, and the final model (trained with the optimized hyperparameters) then applied to the validation cohorts for reporting results.

4. Further on the above, a nested 5-fold cross-validation would possibly yield 5 different sets of optimized hyperparameters (1 for each fold). How are these multiple sets of hyperparameters used to generate a single set of hyperparameters, to train the final model? Or is the final model an ensemble?

5. In the response to Point 8, it is clarified that the DeLong test was used to optimize the number of input features. It might be clarified as to whether the AUC used for this optimization was computed on the discovery cohort (possibly as part of cross-validation), or directly on the validation cohort(s).

(Remarks on code availability)

Code appears to be largely a set of standard machine learning algorithms, as reflected by the authors in their response.

Reviewer #2

(Remarks to the Author)

The authors have answered comprehensively to the criticisms raised in my review, I have no further comments to add.

(Remarks on code availability)

I have installed and run the code which is a usable resource for the community

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

We thank the authors for fully addressing our remaining concerns.

(Remarks on code availability)

The GitHub code appears to be standard implementations of various common machine learning models in Python/SkLearn.

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Reviewer #1 (Remarks to the Author):

This manuscript describes the development and validation of a random forest (RF) machine learning-based screening model, for the prediction of primary vitreoretinal lymphoma (PVRL) from six input features derived from complete blood count (CBC) data. The RF model achieved AUC=0.85 in the discovery/test set, and was further evaluated on a large prospective cohort of some 100,000 subjects, and a community cohort of over 500,000 subjects.

The potential of an accurate noninvasive screening tool for PVRL from commonly-encountered CBC data is of interest, especially given the lack of a standardized diagnostic protocol for PVRL. Some issues might be considered:

1. In the Introduction section, it is claimed that there is no standardized diagnostic protocol for PVRL. The choice of the 2016 WHO classification (as described in the Diagnostic Criteria for PVRL subsection in Line 147) might thus be clarified further. Is the 2016 WHO classification biopsy protocol considered the gold standard for PVRL diagnosis then?

Response: We sincerely thank the reviewer for this valuable and constructive comment, which has helped us improve the clarity and accuracy of our manuscript. We agree that our previous statement ("Currently, there are no standardized diagnostic protocols for PVRL") was inaccurate. Histopathological biopsy criteria, as outlined in the 2016 WHO classification, are currently considered the gold standard for confirming PVRL.

We have revised the sentence to: *Although standardized diagnostic protocols for PVRL exist, the rarity and fragility of lymphoma cells in the vitreous often hinder prompt detection, making timely diagnosis challenging.*

2. In the Study design and participants subsection, it is mentioned that healthy controls were recruited from the health examination centres of various hospitals. It might first be clarified as to whether these were the same hospitals that the PVRL patients were recruited from. Then, since the study is described as case-control, it might be clarified as to how controls were

matched to cases (e.g. by age, gender, etc.), and whether such matching was one-to-one or one-to-many.

Response: We thank the reviewer for this insightful comment. Healthy controls were recruited from the health examination centres of the same hospitals where the PVRL patients were enrolled. In this study, the healthy controls were matched to cases by age and gender. A one-to-one matching strategy was applied to ensure comparable demographic characteristics between groups. As shown in Tables S2 and S3, there were no statistically significant differences in age or gender between the normal control and PVRL groups ($P > 0.05$). We have incorporated this information into the revised Study design and participants subsection to clarify the study design.

3. Related to the above, a flowchart detailing the recruitment and exclusion for both cases and controls for the case-control multicentre study should be included if possible (as was done for the external cohorts in Figure 5).

Response: We sincerely thank the reviewer for this valuable comment. A flowchart detailing the recruitment and exclusion process for both cases and controls in the case-control multicentre study has been added to the revised manuscript as new Figure S1.

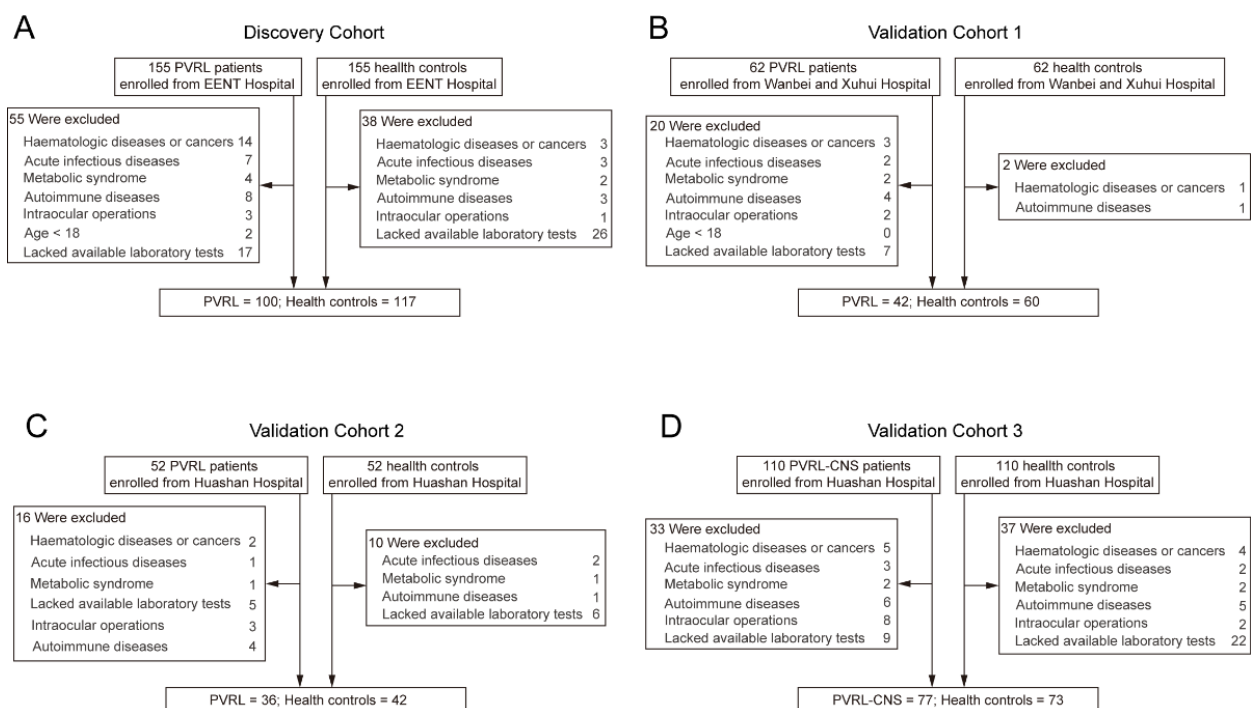


Figure S1. Flowchart illustrating the recruitment and exclusion criteria of cases and controls in the multicentre case–control study across the discovery cohort (A), validation cohort 1 (B), validation cohort 2 (C), and validation cohort 3 (D). The diagram summarizes the numbers of participants initially assessed for eligibility, the reasons for exclusion at each step, and the final numbers included in the analysis for both PVRL cases and control groups.

PVRL: primary vitreoretinal lymphoma; PVRL-CNS: PVRL patients with central nervous system involvement.

4. In the Study design and participants subsection, it is then stated that PVRL screening was restricted to individuals over 40 years of age, to enrich prevalence within the study cohort. However, given that PVRL has extremely low prevalence to begin with, would this not unnecessarily limit the applicability of the developed model for the younger (<40 year-old) population? This exclusion criteria might thus be further justified.

Response: We thank the reviewer for this valuable comment. In both the hospital-based prospective cohort study and the community-based cross-sectional screening program, PVRL screening was restricted to individuals over 40 years of age. This decision was made for two main reasons. First, due to the low prevalence of PVRL and its higher incidence in the elderly population, restricting the age range allowed for a more efficient evaluation of our model's performance. Second, without this age restriction, approximately 1.5 million individuals would have needed to be screened, which would have rendered the study logistically unfeasible given the available resources. Accordingly, 100,526 participants were recruited into the hospital-based cohort and 515,326 participants were recruited into the community-based cohort.

It is important to note that in the discovery phase of the case-control study, during which the screening model was developed, the inclusion criterion for age was ≥ 18 years (as detailed in the Supplementary Methods). The model developed in this discovery cohort demonstrated good screening performance in both validation cohort 1, validation cohort 2 and validation cohort 3, where the same age criterion (≥ 18 years, detailed in the Supplementary Methods)

was applied.

Therefore, while the hospital-based and community-based screening phases restricted PVRL screening to individuals over 40 years of age for practical and epidemiological reasons, this does not limit the theoretical applicability of the developed model to younger (<40-year-old) populations.

Nevertheless, we acknowledge the limitation that the applicability of the developed model to younger (<40-year-old) individuals has not been extensively validated in large population-based cohorts. This limitation has now been noted in the Discussion section of the revised manuscript.

5. In the Diagnostic Criteria for PVRL subsection, it is stated that "PVRL was diagnosed based on a positive pathological finding from vitreous biopsy". Any detailed methodology and quantitative criteria leading to said positive pathological finding, might be described.

Response: We thank the reviewer for this valuable comment. In our study, PVRL was diagnosed based on a positive pathological finding from vitreous biopsy. Undiluted vitreous samples were obtained via dry vitrectomy and processed for cytological analysis. A positive pathological finding was defined as follows: conventional smear cytology demonstrating large lymphoid cells with irregularly shaped nuclei, multiple prominent nucleoli, and scant basophilic cytoplasm, typically accompanied by small reactive T lymphocyte infiltration, consistent with large lymphomatous cells.

These methodological details have been added to the Methods section of the revised manuscript.

6. In the Model development subsection, it is stated that 29 input features were available to develop the diagnostic models. These 29 features might be provided, possibly in the supplementary material (are they in Table S2)?

Response: We thank the reviewer for this helpful comment. We would like to apologize for

the typographical error in the manuscript. The actual number of input features used is 30, rather than 29 as originally stated. This correction has been made consistently throughout the manuscript following a thorough review.

As noted, the 30 input features used for model development are detailed in the Supplementary Material (Complete Blood Count section) and in Table S2. For clarity, we have also explicitly listed them in the Model Development subsection of the main text.

The 30 features were: neutrophil count, neutrophil percentage, red blood cell count (RBC), thrombocytocrit (PCT), platelet count (PLT), platelet distribution width (PDW), haemoglobin (Hb), eosinophil count, eosinophil percentage, basophil count, basophil percentage, mean platelet volume (MPV), lymphocyte count, lymphocyte percentage, haematocrit (HCT), monocyte count, monocyte percentage, platelet large cell ratio (PLCR), white blood cell count (WBC), red blood cell distribution width-standard deviation (RBCSD), red blood cell distribution width-coefficient of variation (RBCCV), mean corpuscular volume (MCV), mean corpuscular haemoglobin concentration (MCHC), mean corpuscular haemoglobin (MCH), platelet-to-lymphocyte ratio (PLT/lymphocyte count), neutrophil-to-lymphocyte ratio, lymphocyte-to-monocyte ratio, $PLT \times$ neutrophil-to-lymphocyte ratio, $PLT \times$ neutrophil $\times$ monocyte-to-lymphocyte ratio, and neutrophil $\times$ monocyte $\times$ PLT-to-lymphocyte ratio.

7. In the Model development subsection, it is stated that grid search and manual finetuning were used to determine the hyperparameters for each model. From the supplementary material, 5-fold cross-validation was performed (Figure S1), which implies possibly 5 different sets of optimal hyperparameters (one for each fold). It might thus be further clarified as to how the hyperparameters were derived (e.g. are the 5-fold results shown, from the set of hyperparameters with the optimal mean performance over all folds?)

Response: We thank the reviewer for this helpful comment. We confirm that a single robust set of hyperparameters was applied to each algorithm, rather than fold-specific hyperparameters. In other words, one optimal hyperparameter configuration was used consistently across the 5-fold cross-validation. Hyperparameter selection was performed in two stages:

First, a grid search with nested 5-fold cross-validation was conducted for each algorithm. A discrete grid of candidate hyperparameter combinations was predefined, and each combination was evaluated using 5-fold cross-validation, where in each fold the model was trained on 80% of the data and validated on the remaining 20%, yielding five performance scores per combination.

Second, the mean-optimal configuration was identified by calculating the mean area under the curve (AUC) across all folds for each candidate hyperparameter combination. The configuration achieving the highest mean AUC was selected as the final hyperparameter set. Minor manual fine-tuning (e.g., adjustment of learning rate or max_depth) was performed only around the grid-derived optimum to enhance stability. The final selection criterion remained the highest mean performance from the 5-fold cross-validation, thereby ensuring the adoption of a robust and generalizable hyperparameter configuration.

Minor manual fine-tuning (e.g., adjusting learning rate or max_depth, as detailed in Table S1) was performed only around this grid-derived optimum to enhance stability. The principal selection criterion remained the highest mean 5-fold cross-validation performance, thereby ensuring one consistent and generalizable hyperparameter set for each algorithm.

8. In the Feature selection and model explanation subsection, it is stated that sample features were selected in the discovery cohort with SHAP, and this narrowed the model from 29 features to 1 feature. However, the abstract claims that the RF model includes 6 features (also in Figure 2). The claim of 1 feature might thus be clarified.

Response: Thank you for pointing out this ambiguity. We will revise the text to clarify the feature selection process as follows:

We first ranked all 30 features in the discovery cohort according to their mean SHAP values. Next, we evaluated the predictive performance of models built with the top-ranked features in a stepwise manner—starting from the single most important feature and incrementally adding features (top 1, 2, 3, ..., up to 30)—and compared each model to the full 30-feature model using the DeLong test.

As shown in Figure 2A, the RF model using the top 6 features achieved an AUC slightly higher than that of the full 30-feature model. In contrast, the model with only the top 5 features showed a significant decrease in AUC ($p < 0.05$). Therefore, our final RF model retained these 6 features to balance parsimony and predictive performance.

The current text stating that the model was narrowed “to 1 feature” will be corrected to reflect this stepwise selection process and the final inclusion of 6 features. The revised description has been incorporated into the Feature selection and model explanation subsection of the manuscript.

9. The final choice of 6 features (and not for example 5 or 7 features) might be justified. Was some feature selection methodology applied (e.g. progressively adding features to the models in order of SHAP value, then evaluating the performance on the discovery set)? This might be described in detail.

Response: We thank the reviewer for this important point. To select exactly six features, we applied a sequential forward-selection procedure guided by SHAP importance and statistical evaluation as follows:

Ranking by SHAP importance: A model was trained on the full discovery cohort to compute mean SHAP values for all 30 features, generating a global importance ranking.

Iterative model building: Starting from the top-ranked feature, models were built incrementally by adding one feature at a time in descending SHAP order (top 1, top 2, ..., up to all 30 features).

Performance evaluation: For each model, the AUC was computed on the discovery set using 5-fold cross-validation to stabilize estimates.

Statistical comparison: Each model's performance was compared to that of the full 30-feature model using the DeLong test. As shown in Figure 2A, the RF model using the top 6 features achieved an AUC slightly higher than that of the full model (30 features), while the model with only the top 5 features showed a significant decrease in AUC ($p < 0.05$).

Final selection: To balance parsimony and predictive performance, we retained the top six SHAP-ranked features in the final RF model.

We will update the Methods and Results sections accordingly to detail this stepwise SHAP-guided forward selection, cross-validated AUC assessment, and DeLong test procedure.

10. For the hospital-based and community-based validation, it is assumed that the gold standard biopsy-based ground truth for PVRL was not available for all patients. The analysis of false negatives (i.e. PVRL patients judged to be low-risk and thus not examined in detail, especially for the community-based group) might thus be justified further.

Response: We thank the reviewer for this valuable comment and fully agree that the gold standard biopsy-based ground truth for PVRL was not available for all subjects. Consequently, false negatives—patients with PVRL who were classified as low-risk by our model—may exist, especially in the community-based validation cohort. This represents a limitation of our study.

We have acknowledged and discussed this limitation in the Discussion section of the revised manuscript, which now includes the following text:

“Some patients classified as low risk by our model did not undergo gold standard biopsy-based confirmation of PVRL status, particularly in the hospital-based and community-based validation cohorts. Consequently, some patients categorized as low risk may in fact have had undiagnosed PVRL, representing potential false negatives. This inherent limitation in diagnostic verification could have led to an underestimation of the true prevalence of PVRL and may have affected the reported performance metrics of our predictive model.”

11. The GitHub code appears to be standard implementations of various common machine learning models in Python/SkLearn.

Response: Yes, that is correct. We based our work on standard implementations of common machine learning models in Python’s scikit-learn library. The submitted code includes five-fold cross-validation implementations of 12 baseline machine learning models, with complete documentation of the model development workflow, hyperparameter settings, and performance metric outputs. In addition, it provides model interpretation (including SHAP plots). The GitHub repository also contains the external validation workflow and the source code for a web-based prediction tool developed with Streamlit.

Reviewer #2 (Remarks to the Author):

This interesting study aims to develop a machine learning–based screening model for detection of PVRL using particular selected complete blood count parameters, i.e., monocyte percentage, Platelet Large Cell Ratio (PLCR), monocyte count, HG and basophil count. I congratulate the authors for their very thorough study, demonstrating multiple validation levels across both inpatient and outpatient settings. The results show that a six-feature-based random forest machine learning model is capable of predicting patients with vitreoretinal lymphoma. Furthermore, the study also mentions a comparison of patients with VRL with those with uveitis and vitreous opacities. It would be very interesting to know the specific diagnostic entities for the latter, because it's one thing to consider easily differentiated clinical uveitis (herpetic uveitis), but it's quite another to consider uveitis from sarcoidosis or endogenous endophthalmitis as a differential diagnosis. For these patients, it has been demonstrated that radiomic analysis of anterior segment OCT images is a non-invasive diagnostic aid (Gozzi et al. Diagnostics 2023).

One aspect that needs to be emphasized, however, is that this machine learning method based on these CBC parameters must ALWAYS be confirmed by a biopsy, i.e., vitrectomy.

Response: We sincerely thank the reviewer for the positive evaluation of our work. We fully agree with your valuable comment that it is important to clarify the specific diagnostic composition of the control group, as including easily distinguishable cases of typical uveitis (such as herpetic uveitis) is one thing, whereas including uveitis caused by sarcoidosis or endogenous endophthalmitis as a differential diagnosis is quite another. In our study, among the 158 patients with uveitis, 132 had endogenous, non-infectious causes, such as Fuchs uveitis, sarcoidosis uveitis, and Behçet uveitis; 26 cases had exogenous, infectious causes, including bacterial endophthalmitis–associated uveitis (n = 7) and herpetic uveitis (n = 19). This information has been added to the revised manuscript.

Interestingly, Gozzi et al. developed a classification model using Python and XGBoost; in our model, 87% of eyes were correctly diagnosed as VRL or uveitis (including Fuchs uveitis, sarcoidosis uveitis, Behçet uveitis, and uveitis of unknown origin). Consistent with Gozzi et al. (Diagnostics 2023), who demonstrated that radiomic analysis of anterior segment OCT

images can non-invasively distinguish PVRL from uveitis of various etiologies, our findings further support the potential of machine learning–based approaches in assisting the differential diagnosis. This point has now been incorporated into the discussion section.

Finally, we fully agree with your important remark that any machine learning method based on CBC parameters must always be confirmed by biopsy, i.e., vitrectomy. This clarification has been explicitly added to the Discussion section, which now reads as follows: “By integrating machine learning with routine blood tests, this study proposes a novel, noninvasive, and high-accuracy screening tool for PVRL. Nevertheless, it should be emphasized that any machine learning approach based on CBC parameters for PVRL screening must be confirmed by positive pathological findings from a vitreous biopsy. In clinical practice, such a tool has the potential to markedly reduce diagnostic delays, facilitate timely intervention, and ultimately improve outcomes in patients with this aggressive malignancy.”

Reviewer #2 (Remarks on code availability): I was not able to install and run the code

Response: We have provided the complete code in our GitHub repository. In brief, to run the code, the user only needs to install Python (versions 3.9–3.11), create an isolated virtual environment, and install the dependencies listed in requirements.txt. The scripts in the master branch can then be used to perform five-fold cross-validation training and evaluation for 12 machine learning models, conduct external validation, and generate SHAP interpretation plots. We have also included a Streamlit application script for deployment (which can be launched using `streamlit run app.py`). The README file provides detailed instructions on the required input data format, example commands, and troubleshooting steps for common issues.

Reviewer #1 (Remarks to the Author):

We thank the authors for addressing our previous comments. Some minor issues might be considered:

Response: We sincerely thank the Reviewer for the careful evaluation of our manuscript and for acknowledging our previous revisions. We are grateful for your constructive and insightful comments, which have further helped us enhance the clarity and rigor of our work. All minor issues raised have been carefully considered and addressed in the revised manuscript, as detailed in our point-by-point response.

1. In the response to Point 2, it is confirmed that a one-to-one matching strategy was applied. In this case, the range for age being matched on might be stated - did the matching have to be exact, or within e.g. 2 years?

Response: We thank the reviewer for this insightful comment. In our study, a strict one-to-one exact matching strategy was applied for age, meaning that cases and controls were matched on the same age in years without allowance for a range (e.g., ± 2 years). This approach was chosen to minimize potential residual confounding related to age. Accordingly, the revised sentence in the Methods section now reads:

“A one-to-one exact matching strategy was applied, with cases and controls matched on age and sex, to ensure comparable demographic characteristics between groups and to minimize potential confounding effects.”

2. In the response to Point 5, it might be briefly clarified as to whether the diagnosis of PVRL from vitreous biopsy followed any established standards, with appropriate references if available.

Response: We thank the reviewer for your comment. In this study, PVRL was diagnosed based on positive pathological findings from vitreous biopsy, following established standards for vitreous sampling and biopsy-based diagnostic criteria [1,2]. The relevant references have been added to the revised manuscript.

[1] Primary vitreoretinal lymphoma: a diagnostic and management challenge. *Blood* 2021; 138: 1519–1534.

[2] Analysis of intraocular biopsies. *Dev Ophthalmol* 2012; 49: 96–116.

3. In the response to Point 7, it is stated that nested 5-fold cross-validation was conducted. It might be clarified whether this nested cross-validation was performed only on the discovery cohort, and the final model (trained with the optimized hyperparameters) then applied to the validation cohorts for reporting results.

Response: We thank the reviewer for this valuable comment. As noted, the nested 5-fold cross-validation was performed only on the discovery cohort, and the final model (trained with the optimized hyperparameters) was subsequently applied to the validation cohorts for reporting results. The revised sentences are as follows:

“First, a grid search was used to define multiple candidate hyperparameter configurations. A nested 5-fold cross-validation was then performed for each candidate configuration in the discovery cohort.”

“The final model, trained with the optimized hyperparameters, was then applied to the validation cohorts to generate the reported results.”

4. Further on the above, a nested 5-fold cross-validation would possibly yield 5 different sets of optimized hyperparameters (1 for each fold). How are these multiple sets of hyperparameters used to generate a single set of hyperparameters, to train the final model? Or is the final model an ensemble?

Response: We sincerely thank the reviewer for this valuable and constructive comment, and we apologize for the ambiguity in our previous response.

First, the discovery cohort was evenly split into five folds, with one-fold used as the validation set and the remaining four as the training set.

For hyperparameter selection, multiple candidate hyperparameter configurations were first defined using a grid search. Each candidate configuration was applied consistently across all five folds, meaning that the cross-validation was used to evaluate the performance of pre-specified hyperparameters rather than to generate different hyperparameters for each fold. The same hyperparameter settings were used to train the model on each of the five training folds and to evaluate it on the corresponding validation fold. As a result, five AUC values were obtained for each candidate configuration, which were then averaged to assess its overall performance. The configuration with the highest mean AUC was selected as the final hyperparameter set for training the final model. The final model is trained using this single best configuration and is not an ensemble of multiple models.

For example, suppose one candidate configuration is learning rate = 0.01 and max depth = 5. The model is trained five times, once per fold, always using this same configuration. The resulting five AUCs are averaged, and this average is used to compare with other candidate configurations.

We have updated this part of the methodology to prevent any ambiguity or misunderstanding.

5. In the response to Point 8, it is clarified that the DeLong test was used to optimize the number of input features. It might be clarified as to whether the AUC used for this optimization was computed on the discovery cohort (possibly as part of cross-validation), or directly on the validation cohort(s).

Response: We thank the reviewer for this comment. The AUC used to optimize the number of input features was computed on the discovery cohort, as part of the cross-validation process. This clarification has been added to the revised manuscript.

Reviewer #2 (Remarks to the Author):

The authors have answered comprehensively to the criticisms raised in my review, I have no further comments to add.

Response: We sincerely thank the reviewer for your careful evaluation and constructive feedback. We are grateful for your positive comments and appreciate your acknowledgment of the revisions.